Capstone Project Weekly Report

Reporting Week:	Week ending October 04, 2026	**Sponsor:**	Deloitte
Project Title:	Air Conditioning Demand Forecasting & Supply Chain Optimization	**Report Owner:**	Vani Som (PGID: 12510023)
Team Members:	Arvind, Hariharan, Vani, Vedika, Sai Prashanth	**Program:**	ISB AMPBA (Summer 2026)

Project Milestones

The team successfully designed and implemented the Business Decision Intelligence and Inventory Optimization Layer, bridging predictive machine learning models with actionable supply chain execution. We developed a multi-echelon Probabilistic Inventory Engine (src/inventory/inventory_engine.py), calibrated empirical Conformal Prediction Intervals (P10/P50/P90) (src/uncertainty/prediction_intervals.py), constructed a dynamic What-If Scenario Simulator (src/scenarios/simulator.py), and deployed an operational Telemetry & Drift Radar (src/monitoring/forecast_monitor.py, src/monitoring/drift.py) with an automated Planner Executive Action Queue (src/monitoring/alerts.py).

Progress Made in Current Week

1. Probabilistic Inventory Optimization Engine

- **Definitive Replenishment Formulations:** Implemented src/inventory/inventory_engine.py translating SKU-level demand distributions into definitive replenishment policies: Lead Time Demand (LTD = forecast demand over the next L weeks), Safety Stock (SS = Z * sqrt(L) * sigma), and Reorder Point (ROP = LTD + SS).
- **Empirical Variance Derivation:** Derived the weekly error dispersion sigma from the champion's out-of-sample walk-forward backtest residuals (falling back to backtest RMSE), so sigma_LTD = sqrt(L) * sigma reflects measured forecast error rather than raw demand variability.
- **Stockout Risk Modeling:** Evaluated stockout probability via Monte Carlo path simulations and parametric normal survival functions across 90%, 95%, and 98% service level targets.
- **Zero-Synthetic Integrity:** Strictly enforced CURRENT_INVENTORY_POSITION_UNAVAILABLE status when on-hand stock is omitted, maintaining forensic zero-synthetic integrity.

2. Uncertainty Quantification & Conformal Prediction Intervals

- **Distribution-Free Conformal Intervals:** Developed src/uncertainty/prediction_intervals.py providing distribution-free split-conformal prediction intervals, level-scaled from the champion's out-of-sample backtest residuals so they widen in peak season and narrow in the trough.
- **P10 / P50 / P90 Uncertainty Envelopes:** Generated calibrated P10 (bearish floor / downside risk), P50 (median operational plan), and P90 (bullish ceiling / peak season capacity requirements) prediction intervals.
- **Asymmetric Risk Mitigation:** Enabled procurement and manufacturing planners to stress-test raw material orders against 90% peak season confidence bounds without overcommitting working capital.

3. What-If Scenario Stress Testing & Simulation Engine

- **Dynamic Operational Stress Tests:** Built src/scenarios/simulator.py to evaluate supply chain resilience under planning-assumption demand scenarios (Summer Heatwave +30%, Dealer Festive Scheme +20%, Supply Chain Headwind -20%, Aggressive Market Expansion +15%, or a custom multiplier) with user-set lead time and service level.
- **Working Capital & Stockout Exposure:** Recomputes safety stock, reorder point and safety-stock capital under each scenario for any of the 5 regional sales branches; stock-out risk is reported only when the user enters on-hand stock.

4. Live Telemetry, Statistical Drift Radar & Planner Alert Queue

- **APICS Tracking Signal Telemetry:** Developed src/monitoring/forecast_monitor.py tracking cumulative tracking signals (TS = sum(e_t) / MAD) on the champion's latest out-of-sample backtest fold against standard APICS quality control boundaries (+/- 4.0 'Watch', beyond +/- 6.0 'Retrain Required' flag), together with accuracy degradation versus the model's earlier folds.
- **Statistical Drift Radar:** Implemented src/monitoring/drift.py calculating Population Stability Index (PSI) with quantile-based binning and two-sample Kolmogorov-Smirnov (KS) tests to detect demand regime shifts.
- **Executive Action Queue:** Built src/monitoring/alerts.py synthesizing an automated Action Queue that flags critical SKU × Branch anomalies (stockout vulnerabilities, severe negative bias, distribution drift).

Business & Model Observations

- **Working Capital Sizing from Empirical Error:** Safety stock is sized as SS = z * sqrt(L) * sigma, with sigma taken from the champion's out-of-sample backtest residuals rather than from raw demand variability, and ROP = LTD + SS; the buffer therefore reflects how well each SKU × Branch is actually forecast. No buffer-reduction percentage has been measured.
- **Lead-Time and Branch Sensitivity:** The simulator recomputes LTD, safety stock and ROP for any lead time and service level per branch (ODU volume share: MAA 35.3%, SBD 28.0%, SBD1 16.1%, COK 10.8%, BLR 9.8%); stock-out probability requires the

planner to enter on-hand stock, which is not in the billing extract.

- **Drift and Degradation Monitoring:** The tracking signal and accuracy degradation are computed on the champion's latest out-of-sample backtest fold relative to its earlier folds, and PSI / KS tests check for drift in weekly demand; series breaching the thresholds are flagged in the planner action queue for review.

Updates / MoM from Sponsor & Faculty Mentor

Deloitte Sponsor Updates:

- Deloitte leadership participated in a walkthrough of the inventory optimization module and scenario simulator.
- Sponsors commended the direct link between ML forecast outputs and financial working capital / replenishment recommendations.
- Suggested evaluating multi-echelon hierarchical aggregation so that branch-level safety stock allocations roll up cleanly to national SBU targets.

Faculty Mentor Updates:

- Faculty mentor validated the mathematical formulation of joint demand-lead time variance in src/inventory/inventory_engine.py.
- Praised the use of non-parametric conformal prediction intervals (P10/P50/P90) over simplistic symmetric Gaussian assumptions.
- Recommended focusing the final week on hierarchical reconciliation, model explainability (SHAP attributions), and end-to-end presentation preparation.

Challenges

Technical Challenges:

- Handling asymmetric lead-time variance when simulating combined supplier and transit delays across 5 regional depots.
- Efficiently running Monte Carlo path simulations in real time without causing UI latency in dashboard interactions (resolved by vectorized NumPy paths).

Non-Technical Challenges:

- Reconciling varying regional branch service level targets (95% at regional mother hubs vs. 90% at peripheral depots) with financial carrying costs.

Key Highlighted Action Items & Next Week Tasks

Key Highlighted Action Items:

- Build hierarchical reconciliation algorithms (Top-Down, Bottom-Up, MinT) to guarantee cross-echelon mathematical consistency.
- Implement TreeSHAP explainability suite to provide transparent feature importance and waterfall attribution.
- Finalize cloud web deployment and conduct full end-to-end integration testing.

Tasks to Finish in Next Week:

- Complete Hierarchical Forecasting & Reconciliation (src/hierarchy/aggregation.py, src/hierarchy/reconciliation.py).
- Implement Explainability Suite (src/explainability/shap_analysis.py).
- Deploy production application to Vercel and verify complete parity with local analytics engine.
- Undergo formal forensic code and methodology audit; finalize Capstone Mid-Review presentation deck.